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An efficient artificial neural network-based optimization techniques for the early prediction of coronary heart disease: comprehensive analysis

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Coronary heart disease (CHD) is the world's leading cause of death, contributing to a high mortality rate. This emphasizes the requirement for an advanced decision support system in order to evaluate the risk of CHD. This study presents an Artificial Neural Network (ANN) based intelligent healthcare system to predict the risk of CHD. The proposed ANN model is trained using the Framingham Heart Study (FHS) dataset, which comprises 4240 data instances and 15 potential risk factors. To combat overfitting, the proposed model uses four hidden dense layers with dropout rates ranging from 0.5 to 0.2. Also, two activation functions, ReLU and LeakyReLU, are used in conjunction with four optimizers: Adam, SGD, RMSProp, and AdaDelta to fine-tune the parameters and minimize the loss functions. Moreover, three sophisticated preprocessing methods, SMOTE, SMOTETomek, and SMOTEENN, along with the proposed two-stage sampling approach, are applied to address the target class data imbalance. Experimental results demonstrate that the Adam optimizer coupled with the ReLU activation function and the combined sequential effect of SMOTEENN and SMOTETomek's two-stage sampling approach achieved superior performance. The validation accuracy reached 96.25% with a recall value of 0.98, outperforming existing approaches reported in the literature. The combined effect of approaches will be evidence of the modern healthcare decision-making support system for the risk prediction of CHD.

Keywords Coronary heart disease, Artificial neural networks, Optimization techniques, Adam optimizer, Performance efficiency, Activation functions

The world nowadays is suffering from many types of diseases such as related to heart, viruses, and cancers. But one of the major causes of death worldwide is coronary heart disease (CHD) as it not only kills people but affects most people[1]. CHD can affect a person by narrowing the tiny blood vessels that carry blood and oxygen to the heart. This is brought on by plaque accumulation, which stops the heart's supply of oxygen-rich blood. The artery disease does not prevent blood flow but causes a blood clot which obstructs the transport[2]. A person with CHD not only suffers from the disease but also the whole family of the patient both emotionally and financially[3]. These cardiovascular diseases can lead to not only deaths but also to costly treatment and expensive medication. Hence it is helpful for anyone to know their probability of suffering from CHD in the future based on their certain activities. Coronary heart disease affects 1% of baby births in the United States every and with more than 40,000 people affected every year[4]. The global CHD death was 217,000 in 2019 while there were 50 deaths in 129 populous countries of Asia such as India, China, Pakistan, and Nigeria have the highest death mortality rate of 39.7%[5]. Also, the World Health Organization (WHO) reported that heart attacks and strokes accounted for 32% of all cardiovascular disease-related deaths in 2019[6]. According to data from India's Ministry of Health and Family Welfare, cardiovascular diseases (CVD) account for 28.1% of all fatalities[7,8]. Hence, the early detection of heart disease is essential to minimize the mortality rate.

Globally, the expense of treating cardiac disease is quite high and could surpass $1 trillion by 2035[9]. Between 2017 and 2018, the US spent over $229 billion in a year on it[10]. If one were to concentrate on India, the WHO

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estimates that heart-related illnesses cost the nation $237 billion between 2005 and 2015[11]. Hence, minimization of the treatment cost for HD has become extremely important. To ascertain if a patient can have HD, numerous studies have used classical machine learning techniques[12,13], and[14] based on distinguishing characteristics.

So, CHD appears to be the cause of the elevated death rate. This specific illness requires urgent care. Heart attacks can also result from other heart conditions, such as coronary artery disease[15]. CHD is the leading cause of death in developed nations. Heart disorders are thought to be caused by several risk factors, including diabetes, excessive cholesterol, smoking, and inheritance[16]. Many believe that aging is another factor contributing to heart disease. However, there is no restriction on the age factor for heart disease because a vitamin D deficiency can also cause heart disease in infants[17]. In addition to being essential for early disease detection, symptoms can enhance a patient's course of treatment, level of medical care, and quality of life compared to individuals who receive a late diagnosis[18].

Coronary heart disease (CHD) is one of the most deadly and difficult heart illnesses to prevent in healthy persons among all heart disorders. The most recent study shows that cardiovascular disease caused premature mortality in India to rise to 59%, making it one of the main causes of death. Since cardiac arrest is among the most important issues facing the healthcare industry, defining a novel approach for prediction models is crucial. The major goal of this research is to lower the sudden cardiac arrest mortality rate, which is necessary to stop this disease earlier. Additionally, it lowers the cost of treatment for the traditional diagnosis method, which involves time-consuming blood tests and laboratory investigations. Furthermore, employing traditional machine learning models, such as prediction models based on various classifiers, is less computationally efficient and more likely to overfit or underfit. The success of the different classifiers depends on the examples of data that are provided. Therefore, the alternative strategy makes use of an advanced machine learning technique based on an artificial neural network inspired by the human brain, which can adapt to new knowledge and successfully learn from data, improving performance over time.

These questions can be easily answered and used by the model to predict the health status of the person. Some of the techniques used for this purpose were suggested a long time ago but the recent trends in machine learning (ML) and data mining techniques have led to this research. The increase in the amount of optimization techniques in recent times provides a huge opportunity for promising results and increased accuracy.

To forecast, analyze, and treat CHD, data mining techniques are used, and the results show remarkable accuracy. This effort is still being undertaken in the field. To identify the target class, one can utilize classification techniques like Artificial Neural Network, Deep Learning, Naive Bayes (NB), Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), and Support Vector Machine (SVM), and other classifiers. Deep learning algorithms can be used in many different contexts for prediction. Deep learning is a subset of machine learning, which is a subset of artificial intelligence and can learn from unstructured data. Moreover, deep learning uses a multi-layered computational model to function on multiple levels of abstraction for data representation learning. Deep learning has proved remarkably effective in improving test result accuracy in several fields in recent years, and the neural network is implemented to decide the importance of including image dehazing[19], recurrent spontaneous abortion (RSA)[20] for the optimization of machine learning algorithms, propagation-based stereo image segmentation[21], and also vehicle logo recognition using a deep learning model for the optimization[22].

This study focuses on training the model, ANN is put to work with multiple parameters to increase the efficiency and performance accuracy of the prediction model. Initially, the null values in the collected dataset are examined for removal, and then the data requires an oversampling method as the target labels are unbalanced hence oversampling is employed. The data provided for training is then normalized to save the resources used during the training and is split into two parts for evaluation of the prediction model. The artificial neural network used for the process contains four dense layers with multiple activation functions, combined with multiple optimizers such as Adaptive Moment Estimation optimizer (Adam), Stochastic Gradient Descent (SGD), Root Mean Squared Propagation (RMSProp), and Adaptive Learning Rate Method (Adadelta) for precise predictions. Providing good solutions to optimization is a necessity in ANN-based prediction models. Optimization is employed to fine-tune the weight parameters of each dense layer in the neural network for better model training. This article consists of a deep learning technique for developing a model that is useful in predicting a patient suffering from CHD.

The prime objectives of this study are formulated based on the literature survey and the recent technological developments mentioned in the following way.

- *To develop an intelligent decision-support system for the future risk of coronary heart disease based on Artificial Neural Networks architecture.*
- *To investigate and assess the customized ANN architecture using different optimization techniques and activation functions into dense layers concerning the CHD prediction model.*
- *Moreover, to optimize the proposed ANN models' learning rate, weights, and other parameters will be optimized to reduce the loss function for better performance.*

To fulfill the objectives of this article, the main contributions can be summarized as follows:

- *We design and analyze the fine-tuned customized ANN architecture using different optimization techniques like Adam, SDG, RMSProp, and AdaDelta for the prediction of CHD.*
- *This research study compares and contrasts the effective use of activation functions like ReLU and LeakyReLU in the ANN Dense layers for better outcomes of the neurons.*
- *Also, the optimal values of the tuning parameters like learning rate, epoch, batch size, number of hidden layers, number of neurons in each layer, and momentum are extracted to gain maximum efficiency of the CHD prediction model in terms of validation accuracy, loss function, and other metrics.*

- *Moreover, to address the target class data inequality, the Synthetic Minority Oversampling Technique (SMOTE), STOMETomek, SMOTEENN, and the two-stage sampling approach combined with SMOTEENN and SMOTETomek is applied to control the biased prediction towards a particular class for the highly imbalanced dataset.*

The flow of this study is formulated in multiple sections. Section "Literature reviews" discussed the literature reviews conducted by the other researchers. The background study coupled with ANN and other concepts, is presented in section "Background study". The entire research methodology and measurements are elaborated with the experimental outcomes in section "Materials and methods". Lastly, this study is concluded in section "Conclusion" with the limitations and future scope in section "Limitations and future scope".

Literature reviews

This section aims to cover the literature survey related to coronary heart disease detection and classification using machine learning approaches. One major part of defining the objective of the research study is to identify the appropriate data mining techniques. There are many kinds of data mining techniques, including classification, clustering, regression, outlier identification, sequential patterns, prediction, association rules, and more[23]. This research focuses on applying prediction data mining techniques to forecast coronary heart disease. Random Forest, Decision trees, Naive Bayes, SVM, regression, and neural networks are the main classification and prediction methods that are accessible for prediction purposes[24].

The entire literature is categorized into three sub-types depicted in Fig. 1. One part considers the classical ML classifiers which are primarily used to detect whether a person has coronary heart disease or not. The second approach covers ensemble techniques like Boosting, Bagging, Stacking, and Voting classifiers for the prediction of CHD. Also, the feature selection methods are considered to address the important attributes that will be factors for CHD prediction. The third approach focuses on the deep learning technique based on Artificial Neural Networks, inspired by the human brain structure to determine the target class of CHD. Also, the Genetic Algorithms and Particle Swarm Optimization-based research article combined with neural network model prediction.

Traditional ML classifiers primarily

This approach aims to discuss ML classifiers' capabilities for diagnosing coronary heart disease and highlights their significance in the classification model.

In the article[25], the author explains the utility of traditional ML algorithms such as SVM, KNN, Random Tree, LR, NB, and Gradient Boost (GB) to predict the CHD. The most sought-after performance matrices such as accuracy, specificity, sensitivity, and ROC are employed for evaluation. The dataset is procured from a couple of Hospitals in Nigeria's Kano that contain 506 data points and none includes a null value. The model gives an astonishing accuracy of 92.04% and 90.90% with RF and GB algorithms, respectively. The authors[26], develop a model using a pioneer ML method which uses optimization techniques to improve accuracy. Advanced algorithms such as Firefly optimization are utilized to select the data points and Principal component analysis (PCA) a popular analytical tool is employed to build an efficient system. The dataset is preprocessed and scaled using normalization tools for feature extractions. The model developed is a hybrid created using the DT, PCA, and Firefly optimization tools. The performance achieved with and without optimization shows an increase of more than 20-30% each for recall, precision, and accuracy. The highest accuracy achieved with the optimization technique is 93% of the proposed model.

Also, the article[27], described a creative approach for the effective detection and classification of coronary artery diseases (CAD) with the help of a normalization method combined with good optimization measures. The dataset is preprocessed and the important features are selected using cross-validation and training with the help of Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) algorithms. Several popular algorithms such as KNN, Gaussian NB, Linear Discriminant Analysis (LDA), RF, Multi-Layer Perceptron (MLP), Reg Log, and QDA while variations of SVM namely C-SVC, nu-SVC, and Linear SVM have been wielded by the authors. The accuracy scores achieved with the nu-SVM combined with GA obtained the highest accuracy of 93.08% and 93.07% for testing and training respectively. In,[28], the author talks about solving problems with the help of new technologies such as ML and Big Data analytics to predict the risk of a person suffering from a CHD soon. The Framingham heart study dataset is cleansed and processed by increasing the randomness in the training data using k-fold cross-validation. Popular ML algorithms namely RF, DT, and KNN are used for model training. After applying the feature selection and elimination, the average accuracy for RF is 96.80% whereas DT and KNN achieve an average accuracy of 92.45% and 92.81% respectively. The article[29] proposed an evolutionary

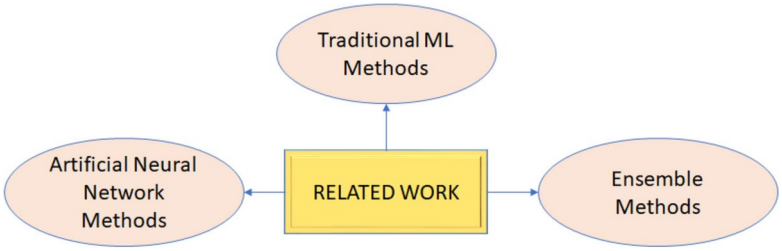


Fig. 1. Literature survey approaches of the classification of CHD.

classification model based on kernel extreme learning machine (KELM) for predicting the CHD. The core model is optimized to get a better subset of features and parameters using the improved salp swarm algorithm (SSA). Also, the space transformation mechanism is used to improve the HELM model for optimum results. The achievable accuracy was observed at 84.40% on the collected hospital dataset with fewer instances for the binary classification model.

Moreover, in the study, the authors[30], developed a prediction model using the Framingham dataset consisting of 4238 rows and 14 columns of data instances. For preprocessing, data handles missing values with the help of the MICE method and SMOTE for balancing the classes with the help of Random sampling to prevent any over-fitting. For improvement of accuracy and efficiency, the Recursive Feature Elimination (RFE) and Boruta are wielded by the author for feature selection. The accuracy achieved by the RF is the highest with the RFE and SVM combined with Boruta being the maximum of 89% and 86% respectively. In[31], the author manifests the essence of a Coronary Artery Classification Score (CACS) which is estimated using Coronary Artery Computed Tomography Angiography (CCTA) and then its analysis by ML algorithms. The dataset that is employed for training the model contains the CACS of the patients admitted to a hospital in China. A few popular categorization algorithms are used to train the model namely RF, SVM, Kernel Ridge Regression (KRR), Radial basis Function Neural Network (RBFNN), and KNN. These algorithms are now employed for judging the risk of a person suffering from a heart disease. Amongst these techniques, RF clinches the highest accuracy of about 78.96%. The authors[32], compare the outcomes of multiple ML algorithms for the classification of CHD and the results achieved from a couple of statistical software's such as R-Studio and RapidMiner. Framingham database is considered and multiple libraries of the software R-Studio such as MASS and ROSE have been employed for data pre-processing. The authors have also worked on data balance and normalization to increase efficiency and accuracy in the model's compilation time. The highest accuracy obtained by the Boosted Decision Tree (BDT) when combined with R-Studio is 85% whereas the DT together with Rapid Miner also achieved 85%.

The selection of machine learning classifiers is a challenging task to perform the CHD prediction model among the diversified algorithms. Also, this approach is computationally expensive, and task-specific outcomes can be observed with lesser performance. Moreover, the generalized model prediction can not the achieved due to data dependency, black-box nature, and more prone to overfitting nature. These limitations and research gaps can be addressed by considering the advanced matching learning approaches.

Ensemble techniques primarily

The ensemble methodology is a data mining method that efficiently combines the best ML classifiers for disease identification. The purpose of this section is to discuss ensemble learning techniques for coronary heart disease diagnosis, specifically stacking, voting, bagging, and boosting.

The authors, in article[33], focus on the prediction of a person suffering from CHD with the help of a risk factor. Algorithms such as KNN, Binary Logistic Classification, and NB are used to train the model on the given dataset for comparison with advanced algorithms such as boosting, bagging, and stacking. The dataset contains more than 70,000 instances with more than 18 columns which have been normalized as they have a huge standard deviation. For analysis of each of the columns to understand the relationship between them Pearson's and LASSO's Coefficient is implemented. Traditional classification algorithms have an accuracy capped at 74% with MLP being the maximum 73.9%. The modern bagging and boosting methods combined with these algorithms are at a limit of 75% and Bagged DT has the highest 74.8% accuracy. In[34], the authors showcase the importance of primitive bagging algorithms that can be essential for the prediction of coronary heart diseases (CHD). The Framingham Heart Study dataset contains instances of more than 3000 patients divided into two parts and their parameters collected over 10 years. The accuracy achieved for XGBoost is 87.62% which is the maximum while the succeeding AdaBoost and Cat Boost obtain 87.22% each.

Also, in[35], the author used a voting classifier with the help of ensemble techniques that help in the detection of CHD in the early stage. The dataset used is obtained from Framingham Medical Institute, Massachusetts USA, and contains 15 parameters for training and testing with more than 1200 instances. Algorithms such as SVM, KNN, DT, and RF are used after data preprocessing and are evaluated by the Hard voting (HVC) and Soft Voting (SVC) classifiers. The HVC chooses the class with maximum votes among the predictions. The accuracy obtained from the SVC and HVC are 82.5% and 83.2% respectively. Moreover, the author[36], suggested the use of a voting-based system combined with Boosting and Ensemble learning techniques for developing a model to predict if a patient is suffering from CHD or not as early as possible. To improve the accuracy of the model hyper-parameters are maneuvered as well which increases the utility of the voting algorithm significantly. The average accuracy achieved by the RF is the highest at 90.69% before hyper-parameter tuning whereas that of GB, CB, and Hist-GB are the highest at about 92.12%, 92.10%, and 91.80% respectively after optimization. The highest accuracy achieved by the model is the SVc with medial accuracy of 92.26% while the precision, recall, and F1-score are 96%, 87%, and 91% respectively.

The ensemble technique is also more complex and requires high computational resources. The performance depends on the choice of classifiers for the created ensemble pipe or the base learner and meta-learners of the selected ensemble techniques. Due to these practical challenges of development, these techniques can be replaced by using an advanced deep learning-based artificial neural network approach. In this case, the performance is not up to the mark, so there is the possibility to enhance the outcomes in more effective ways.

Neural networks approaches primarily

Another approach for the classification of the target class of CHD in an efficient manner is the neural network. This section combines the approaches of ANN along with GA and PSO, and other proposed methods considered in the kinds of literature.

The authors[37], shed light on the utility of neural networks and use a conventional optimization technique for the prediction of disease. Multiple feature selections are used such as Fisher Score, Relief-F, Minimum Redundancy Maximum relevance, and SVM's influence on weight. The efficiency of the model is improved with the use of an Emotional Neural Network (ENN). The PSO is based on an adaptive neuro-fuzzy inference system (ANFIS) the other algorithm used with the ENN in combination uses a hybrid mechanism of ANN paired with Fuzzy Logic. The results achieved for PSO-ANFIS are satisfactory with accuracy being 70% and the combination of PSO-ENN provides an outstanding 86.33% accuracy. Also, in the study[38], applies three different techniques to optimize performance by using Genetic Algorithms (GA) with feature selection reducing the size of the dataset by 20%. The Framingham's Heart Study Dataset was considered for this case. A Tree-based pipeline optimization tool (TPOT) classifier with hyperparameters was used to increase the efficiency of the developed model. To improve the quality of the data and decrease the run-time, the preprocessed data is scaled and put into three different models. The genetic algorithms involved with the popular algorithms such as DT, RF, Gaussian NB, LR, KNN, and the essential boosting techniques. Random Forest with hyperparameter tuning and tree-based optimization achieved the highest accuracy with 90.7% as compared to 91.6% without feature selection. Moreover, the study in[39], used a simple two-layered CNN model for the classification of CHD with an imbalanced dataset. LASSO is a very popular regression method that is employed for variable selection and regularising by enhancing the accuracy of prediction. It is used for removing the problem of multi-collinearity by removing some variables and eliminating them from the model. Initially, the data after pre-processing is pushed through a dense layer with ReLu, and finally SoftMax activation function for prediction of the class. The accuracy achieved using the proposed model is 82.32%.

Also, in[40], the author substantiated the application of six traditional ML algorithms and developed a model for the prediction of coronary artery diseases (CAD). The model is developed using the Cleveland Heart dataset. Four algorithms achieve a mean accuracy of 85% and ANN acquires an accuracy of 93.03%. In a study[41], the paper exhibits the brilliance of the MLP with the help of a deep learning-based neural network and developing a model with a high-quality dataset acquired from the Framingham Heart Study. The algorithm used is popularly known as a feed-forward neural network as it is a linear classification technique. Before preprocessing the accuracy of the model acquired is 92.87% but the sensitivity of this result is only 50%. The model performs overwhelmingly after the dataset used is preprocessed with feature selection and the accuracy achieved is 96.50% with the sensitivity being more than 91%. Lastly, in[42], the authors proposed an improved ANN based on a sparse autoencoder (SAE) that predicts heart disease. The two stages of the learning approach were utilized to train the data patterns during the training process using an unsupervised neural network. While the ANN was utilized to provide predictions based on the learned records, the sparse autoencoder was utilized to identify the optimal data representation. Also, the Batch normalization was used in conjunction with the Adam algorithm to optimize the SAE. the accuracy achieved by 90% of the test data showed that the model was accurate and relative to several conventional methods of machine learning and artificial neural networks.

In outlining the above three types of literature survey, the concerned authors used different algorithms. Their proposed model with observable experimental outcomes in accuracy is presented in a tabular form for easy reference of the readers in Table 1.

In summary of this section, there are numerous learning strategies available for CHD prediction. Comparing Artificial Neural Networks to existing categorization algorithms, however, reveals that they are more efficient. One of the ANN algorithm's main benefits is its ability to handle challenges of complex issues efficiently. Additionally, the literature study examines a variety of applications in the field of artificial neural networks, confirming the evidence of the versatility and universality of the ANN learning scheme. Its learning efficiency is therefore far higher than that of other classification algorithms. Numerous tests conducted on various commonly used categorization data sets demonstrate that the suggested approach outperforms the most advanced learning techniques currently in use.

In literature, none of the articles tried to optimize the neural network architectures using the different optimizers by considering the tuning parameters like learning rate, weights, number of neurons in each layer, number of hidden layers, and others to reduce the loss function of the network. So, the experimental outcomes in terms of performance accuracy and loss function can be improvised by tuning the parameters still in scope for a better understanding of the neural network model in this case.

This research study concentrates on the efficient use of ANN in a customized way of learning method concerning the objective of the problem statement. Next and rest of the sections related to the proposed methodology based on the detection of coronary heart disease using ANN and the learning approaches.

Background study

Understanding the operation of the deep learning terminologies utilized in the model is crucial to comprehending the overall operation of the model generated in the research. A few key components of the established model must be recognized and comprehended to use all of the strategies. As a result, this section explains how each method operates so that the work covered in the other sections can better connect to its applications.

Artificial neural network (ANN)

ANN is a part of machine learning and an essential part of deep learning algorithms. It is inspired by the structure of the human brain and tries to replicate the methods that the biological neurons work with. ANN contains 3 layers of nodes namely the input layer, single or multiple hidden layers, and output layers. The input layer uses a matrix of the rows and columns which contains values of every data point and is used in the hidden layer for further processing. ANN contains two types of functions namely, Summation Function and Activation Function. The general ANN structure is represented in Fig. 2. For this study, a similar kind of neural network

#Ref	Year	Algorithm(s) used	Proposed method	Dataset used #Instances	Accuracy (%)
Traditional ML classifiers primarily					
[25]	2022	SVM, KNN, Random Tree, LR, NB, GB	RF	Hospital 506	92.04
[26]	2021	Firefly optimization, PCA, DT	Hybrid Model	Z-Alizadeh Sani CAD 303	93.00
[27]	2019	GA, PSO, KNN, GNB, LDA, RF, MLP, Reg Log, and QDA, C-SVC, nu-SVC, and Linear SVM	Nu-SVM with GA	Z-Alizadeh Sani CAD 303	93.08
[28]	2019	RF, DT, KNN	KNN	FHS 4238	92.45
[30]	2022	RF, RFE, SVM, Boruta	RF with RFE	FHS 4238	89.00
[31]	2022	RF, SVM, Kernel Ridge Regression (KRR), KNN Radial Basis Function Neural Network (RBFNN).	RF	Hospital 1502	78.96
[32]	2019	Boosted Decision Tree (BDT), DT, Rapid Miner, R-Studio (MASS and ROSE)	BDT	FHS 4238	85.00
[29]	2020	KELM, salp swarm algorithm (SSA)	KELM	Hospital 443	84.40
Ensemble techniques primarily					
[33]	2021	KNN, Binary Logistic Classification, NB, DT, MLP, Lasso, Bagging	Bagged DT	CVD 70000	74.80
[34]	2019	XGB, AB, CB with Bagging	Bagged XGB	FHS 3000	87.82
[35]	2021	SVM, KNN, DT, RF, Voting (SVC, HVC)	Voting-HVC	FHS 4238	83.20
[36]	2022	GB, CB, Hist-GB, Voting (SVC, HVC)	Voting-SVC	FHS 4238	92.26
Neural networks approaches primarily					
[37]	2020	Fisher Score, Relief-F, Minimum Redundancy Maximum relevance, Emotional Neural Network (ENN), PSO, neuro-fuzzy inference system (ANFIS)	PSO-ENN	Z-Alizadeh Sani CAD 303	86.33
[38]	2021	GA with DT, RF, Gaussian NB, LR, KNN	RF with GA feature selection	FHS 4238	91.60
[39]	2020	Lasso, Majority Voting followed by CNN	MLP	NHANES 37079	82.30
[40]	2021	LR, DT, RF, SVM, KNN, ANN	ANN	Cleveland 303	93.03
[41]	2021	MLP	MLP	FHS 4238	92.87
[42]	2020	ANN based on a sparse autoencoder (SAE)	ANN-SAE	FHS 4238	90.00

Table 1. The summary of the literature survey with observable experimental outcomes.

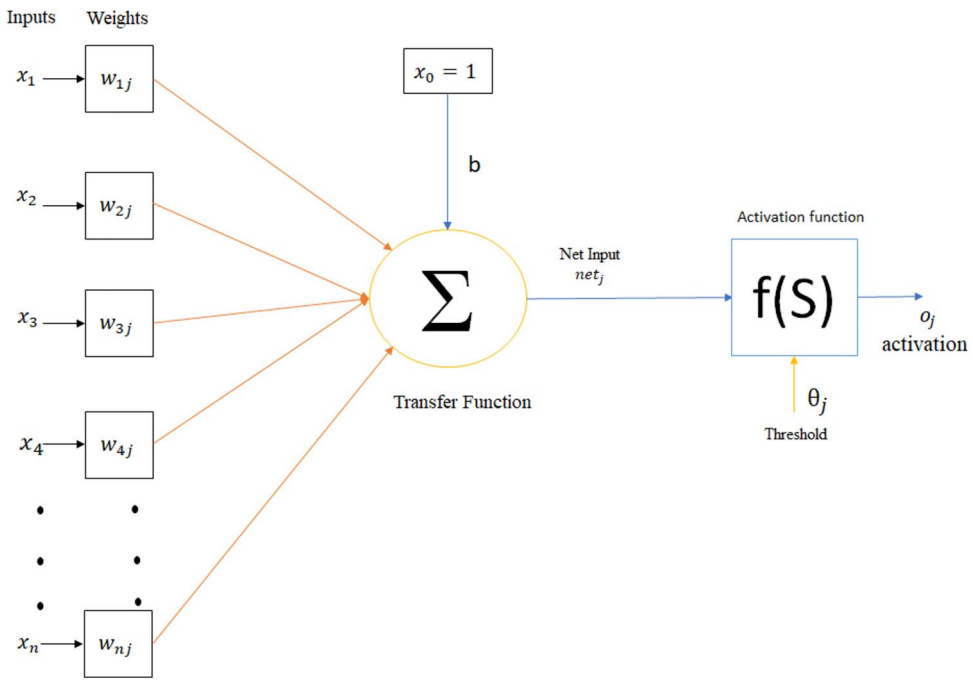


Fig. 2. The general structure of an artificial neural network used in this study.

with one input layer, four hidden dense layers, and one output layer with a single neuron is considered to deploy the CHD model for the binary classification task.

For the input layer the processing is done using the summation function and the output is achieved from the activation function. The following formula is used for achieving the outputs from each neuron by Eq. (1).

$$\sum_{i=1}^{n} w_i x_i + b = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + b \tag{1}$$

where, w_i indicates the weight of every data point, x_i is the input value, and b is called the bias. The output layer uses the given formula in Eq. (2) for achieving the results with the inputs.

$$f(x) = \begin{cases} 1 & \text{if } \sum w_i x_i + b \geq 0 \\ 0 & \text{if } \sum w_i x_i + b < 0 \end{cases} \tag{2}$$

Activation function

This is the most essential component that uses a simple algorithm to decide if the answer is true or false by a neuron. For this, a variety of activation functions are employed like linear functions to predict values with less complexity and don't require significant parameter tuning. It only employs positive and negative signs on values between 0 and 1. Also, the various kinds of non-linear activation functions can employ genuine data value used as the input by the **sigmoid or logistic function**, which predicts a different value between 0 and 1. Any negative input is converted to zero on the graph using **ReLU**, which has an impact on the ANN model. By diverging the value with a very little margin, typically 0.01, the **LeakyReLU** can solve dying neuron problems of ReLU and optimize prediction efficiency by shifting the graph. The probability of an event happening is determined using **SoftMax** by summing the possibilities of each target label across all potential labels. **Tanh** is a function with an output range of -1 and 1 that directly corresponds to the tangent hyperbolic. The Sigmoid Function is a very popular non-linear activation function used in deep learning denoted by **Sigma** (σ) and is used to find the slope of the curve of a graph on any two given points. It is a function lying in between an open interval of (0,1) and is used to fire a neuron while in the middle the slope is large hence, it is a sensitive area of the neuron. It is commonly used for binary categorization in logistic regression models. The Eq. (3) used to find the slope is given below.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

After understanding those summation functions, the task is to the binary prediction of disease, for better bias and weight updating, the non-linear activation function Sigmoid is used in each neuron's output for the next layer. Also in the output layer to correctly predict the disease in binary, the sigmoid function is considered for the proposed model.

Moreover, the *ReLU* and *LeakyReLU* functions are generalized versions of Maxout, which operates based on a linear nature that repeats the highest of inputs. By approaching the normal gradient contiguous, average element activation is shifted closer to zero by ELU's usage of negative values, which acknowledges this shift and accelerates learning. This is also known as the Rectified Linear Unit but is a part of the Non-linear functions group that is used to achieve results from input given to it positive or else zero. It is commonly used in CNN and MLP networks to deal with a specific task. It is very simple in application and yet efficient in comparison to the Sigmoid function. It works on a simple Eq. (4) as given below.

$$f(x) = \max(0, x) \tag{4}$$

Also, for this research, the *LeakyReLU* is considered to much more optimized form of the ReLU activation function and it uses a smaller slope for negative data points alternative to the flat slope represented in Eq. (5). The coefficient of the slope is assumed before training and not estimated during the training. It is commonly used to solve problems with infrequent gradients such as while training generative model networks and dying ReLU.

$$f(x) = \max(0.01, x) \tag{5}$$

Optimization techniques

In deep learning models, algorithms must use the fewest resources possible to save ongoing expenses and deliver effective outcomes faster. An optimizer is a technique for adjusting the different parameters that can lower the loss function with less work. By using an algorithm to generalize the data, all deep learning models forecast data that has not yet been observed. This specific optimization approach maps inputs to outputs while searching the parameter values and minimizing error. Such optimization has a major impact on the deep learning model's accuracy. In this case, this technique is used to train the ANN model multiple times which helps in the evaluation of both maximum and minimum functions also reduces the loss functions, and adjusts weights. So, the optimizer modifies the learning rate and weights of a neural network by fine-tuning it. As a result, it helps to increase accuracy and decrease overall loss. This is done by comparison of the results achieved from the model by using multiple parameters tuning to exact the perfection and pinpoint accuracy. This helps in decreasing the errors of a model's prediction while also keeping in check the fitting of the model with the specified problem.

This process is called backpropagation which is eminent while using ANN. There are multiple ways of achieving this efficiency and a few are discussed below.

SGD (stochastic gradient descent)
This is based on gradient descent that uses calculus to modify the value for achieving the local minimum. To manage the large dataset, pose, SDG used random probability of selecting a few data samples on a random basis which is known as a "batch" and is used for measuring the gradient in each loop. It is a primitive optimization algorithm technique that helps in finding an effective result for multiple problems. It uses a simple principle of tweaking parameters repetitively of unit data value to decrease the cost function and find the most suitable solution. This parameter tweaking is done for every sample of data in each iteration which increases the efficiency of the mechanism. The cost function used for solving the SGD problems is given below in Eq. (6), where, θ: The gradient of cost function, $J(\theta)$: Parameterized objective function, $x^(i)$: Each training examples, $y^(i)$: Target Labels, $\nabla_\theta J(\theta)$: Gradient of the objective function, and η: Learning rate.

$$\theta = \theta - \eta \cdot \nabla_\theta J(\theta; x^{(i)}; y^{(i)}) \tag{6}$$

RMSProp (root mean squared propagation)
Optimizing models for machine learning often involves finding the best weights that minimize errors. Gradient descent is a popular technique for this, but it can be slow and require a lot of calculations. RMSProp is a recent advancement that addresses these issues. Imagine navigating a bumpy landscape to reach the lowest valley (the optimal solution), standard gradient descent takes small steps in the direction of the steepest downhill path. RMSProp improves on this by keeping track of how "bumpy" the terrain has been around each point. It reduces updates for parameters that cause large swings in the error (steeper bumps), making the progress smoother and faster. This stability allows RMSProp to work well even when using smaller batches of data for training. RMSProp builds on earlier ideas like AdaGrad, which unfortunately forgets past "bumpiness" too quickly and adjusts learning rates based on whether errors are consistently increasing or decreasing. RMSProp combines the best of both worlds, providing stable and efficient optimization for deep learning tasks. However, it still relies on manually setting a learning rate, which can be a drawback in some cases.

The algorithm's primary goal is to speed up optimization by reducing the number of function evaluations required to get to the local minimum. The gradient is divided by the square root of the mean squared, which maintains the moving average of squared gradients for each weight. The Eq. (7) defines a value function, denoted by $v(w, t)$ for a system in a state w at a time t.

$$v(w,t) := \gamma v(w,t-1) + (1-\gamma)(\nabla Q_i(w))^2 \tag{7}$$

where, γ: This is a discount factor that balances the immediate reward with the expected future rewards and the higher values give more weight to future rewards. $v(w, t-1)$: Represents the system's value being in state w at the previous time step $(t-1)$. $(1-\gamma)$: This weighs the importance of the immediate reward compared to future rewards. $\nabla Q(w)$: This is a state-value function, which assigns a value to each state w. ∇Q: The exact definition depends on the specific problem. The following formula in Eq. (8) updates the weights.

$$w := w - \frac{\eta}{\sqrt{v(w,t)}} \nabla Q_i(w) \tag{8}$$

AdaDelta optimizer
It is a widely known SGD technique that at its core has an adaptive learning rate in every dimension which helps in solving two major issues. These issues are the continuous decomposition of the learning rate while training and selecting a learning rate for the whole problem manually. It is a very vigorous iteration of the AdaGrad (Adaptive Gradient Descent) method which is known for adapting to the previous gradients and updating the learning rate. Instead, AdaDelta uses a window of gradients to change the learning rate and this improves the technique of optimization and increases the accuracy of the model. It also does not require the learning rate hyperparameter to be constant while solving the problem. In this case, we can use two formulas as follows for AdaGrad represented as in Eqs. (9) and (10), where, t: Time, α: The sum of squared gradients, and η: Learning rate. AdaDelta uses s_{dw_t} in place of α represented in Eq. (11) for small positive constant ε.

$$\eta'_t = \frac{\eta}{\sqrt{\alpha_t + \varepsilon}} \tag{9}$$

$$\alpha_t = \sum_{i=1}^{t} \left(\frac{\partial L}{\partial w_{t-1}} \right)^2 \tag{10}$$

$$s_{\mathrm{dw}_t} = \beta s_{\mathrm{dw}_{t-1}} + (1-\beta) \left(\frac{\partial L}{\partial w_{t-1}} \right)^2 \tag{11}$$

Adam (adaptive moment estimation optimizer)
Adam is a modern and newly developed optimization algorithm that is based on traditional SGD and uses a simple technique of updating network weights iteratively on the training dataset. It differs from the SGD as it

replaces the learning rate during the training and does not maintain a unique learning rate. It uses two other extensions of SGD namely AdaGrad and RMSProp effectively to achieve precise results. It is also popular for achieving staggering results quickly due to the fewer memory requirements in the field of deep learning. It is easy to use as the parameters given to it by default can perform unbelievably well for most of the problems. During training, the Adam optimizer attains an adaptive learning rate that can effectively traverse the optimization landscape by combining the first moment (mean) and second moment (uncentered variance) of the gradients. This adaptivity aids in the neural network's enhanced performance and quicker convergence. The formulas used for updating the weights are given below in Eqs. (12), (13), and (14), such that $\hat{m}_t$ & $\hat{v}_t$: The bias-corrected weight parameters. m_t & v_t: Normal weight parameters. ε: Small positive constant. β_1 & β_2: Decay rates of mean of gradient. α: Step size parameter divided by learning rate.

$$w_{t+1} = w_t - \frac{\hat{m}_t \alpha}{\sqrt{\hat{v}_t} + \varepsilon} \tag{12}$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}^2 \tag{13}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{14}$$

Materials and methods

This section demonstrates the working methodology of the research strategy is to propose ANN-based CHD prediction models using different optimizers. To optimize the neural network, it takes a dataset to efficiently predict the ten years of risk of coronary heart disease of a person. Before deployment of the Neural Network models, the collected raw dataset undergoes preprocessing techniques discussed in a separate section. The deployment models and the evaluation approaches are also mentioned in this part. With the help of this section, the exact details of the models working will be discussed with the different sub-parts that contain the use of the tools discussed in the earlier sections. The working diagram of the proposed artificial neural network-based CHD prediction model is represented in Fig. 3.

Dataset description

The dataset is collected from the HME workshop conducted by the WHO to understand the relationship and predict if the person is suffering from coronary heart disease (CHD) or not[43]. Mortality is due to people suffering from diseases affecting the heart and multiple diseases affecting multiple organs of the human body around the world. The dataset is available publicly for observing patterns and helping predict the probability of a person suffering from CHD. The data is gained from the study based on the people living in the town of Framingham in the state of Massachusetts, USA. The data is collected by reporting the values of more than four thousand patients and fifteen different questions asked from every person. Each question is a very essential factor that can influence the targeted class and all the features are statistical, developmental, and therapeutic. The dataset has

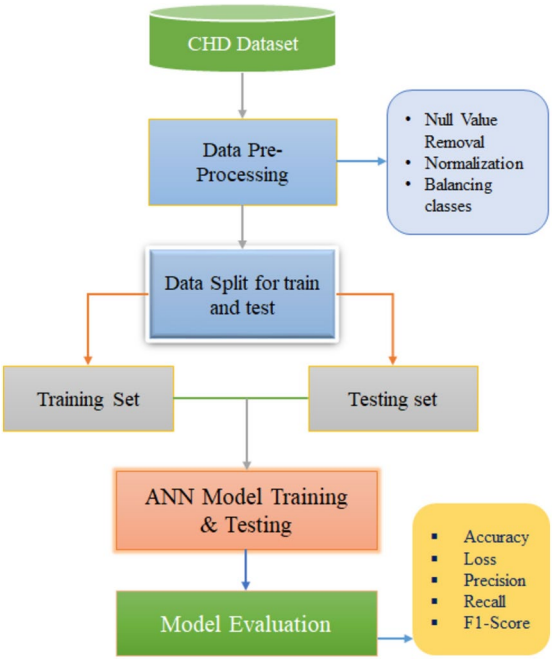


Fig. 3. The workflow diagram of the proposed methodology for CHD prediction.

4240 patients' information summarized into 15 potential risk factors. Each risk factor is categorized into three demographics, behavioral, and medical risk factors with the target variable of 10-year risk of CHD in binary.

Data preprocessing

For the data to be used with good efficacy and precision then it is necessary to process the data. This not only improves the quality of the data to make it more refined and make better decisions. Also, it helps in the model to decrease the running time. The data needs to be scaled to decrease the running time of the model while also removing the null values. One of the primary difficulties in training the dataset is the balancing of the target classes into binary data which is essential. Unbalanced data can cause the model to be unable to have enough evidence to predict a certain class. The data preprocessing methodologies used in this research are given below.

Null values replacement

The most difficult issue in every dataset is the number of null values as it decreases the model's efficacy and reduces its accuracy while also influencing its predicted results. This leads to a major complication and decreases the utility of the dataset. In this context, the missing values impair neural networks by inducing computational mistakes, obscuring patterns, and diminishing accuracy. Inadequate management, such as context-free removal, jeopardizes bias, skews data, and leads to overfitting, hence undermining the model's generalizability and efficacy across varied inputs. Hence, it is necessary to be removed or replaced; for that, it should be detected early. Once the values are detected they are removed by '*fillna*' function and then using the imputation technique to replace the empty values with new values that are the average of values given in the numerical columns. Now it is compulsory to recheck whether all the null values and the missing ones have been eliminated or not. During the data analysis phase, the null values of the dataset were observed in seven attributes with a total of 15.21% missing values. To replace the missing values, imputation methods are used to take the *"mean"* values of each numerical column. Only the "bpmeds" and "education" attribute values are replaced by using the *"mode"* method to handle the categorical feature. Missing data values are depicted in Fig. 4.

Data correlation analysis

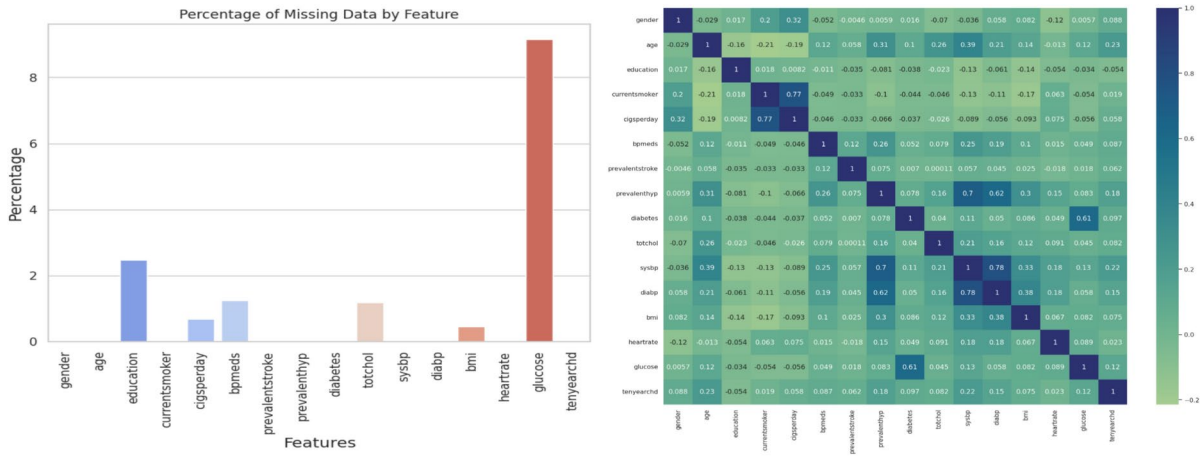
After the necessary steps to replace the null values from the collected dataset, the correlation heat map is generated among the attributes. It is used to identify the linear correlation between the attributes and which one is more potentially multicollinearity with the target attribute. This statically measure also tells the prediction model integrity by verifying each cell's correlated values depicted in Fig. 4. The Pearson correlation coefficient (r) between the two attributes was calculated by the formula below in Eq. (15) in between −1 and 1.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{15}$$

where, r: Correlation coefficient, x_i: Values of the attributes X, $\bar{x}$: Mean of the attribute's values X. y_i: Values of the target class Y, and $\bar{y}$: Mean of the Y values.

Label encoding technique

To achieve high accuracy during training, the desired results must be presented numerically. However, the employment of a categorical datatype is not necessary because the provided data is already in numerical format



(a) Missing values count percentage **(b)** Correlation heat map

Fig. 4. The attributes missing values and correlation heat map among them.

and without any features. Therefore, the label encoder approach is not needed to translate the values of the category columns into numerical form.

Feature scaling or normalization technique

Any model that is deployed during training must be done so on a dataset with little variation, as this can reduce the model's output. Because the values can cause a significant disturbance, this also reduces the accuracy of the model's prediction. To ensure an efficient and effective model, the normalization and scaling used to normalize input features guaranteed uniform contribution, accelerated convergence, and numerical stability in the neural networks model. This approach can optimize the use of activation functions, mitigate the influence of large-scale features, and augment model performance by rectifying feature range discrepancies and outlier impacts. Hence, the data values must be scaled within a range of values between 0 and 1 to increase efficiency. The model's execution time may be shortened with the aid of this normalization and also manages the dataset's outlier. The *"StandardScaler"* function is a mean-based scaling method used for this purpose. It will transform the input data such that its distribution will have a mean value of 0 and a standard deviation of 1. The mathematical formula to transform each variable into scaled values is defined in Eq. (16), where x_i is the instance value of an attribute, x_{mean} is the mean value of the training samples, and σ is the standard deviation of *x* attributes defined in Eqs. (17) and (18).

$$x_{\text{scaled}} = \frac{x_i - x_{\text{mean}}(\mu)}{\text{Standard deviation } (\sigma)} \tag{16}$$

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{17}$$

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2} \tag{18}$$

Balancing target classes

Having adequate data is essential for any model to function properly. Given the binary nature of the dataset, it is essential to divide both classes equally. This leads to a model trained on enough proof to form networks, trees, and graphs as necessary by the ML algorithms. Balancing the target class in a neural network model guarantees equitable treatment of all classes, mitigating bias toward majority classes. This increases the precision of minority class predictions, improves the generalization of the overall model, and prevents biased or incorrect results. A balanced data set fosters equal learning, allowing the model to excel in all goal categories. This can be done using an oversampling technique that can increase the instances of the minority classes with one having fewer than the other. This technique is called the Synthetic Minority Oversampling Technique or SMOTE developed in the year 2002 and is applied to the entire dataset attributes. In the collected dataset, the target class was highly unbalanced and distributed as 644 and 3596 into classes "1" and "0", respectively. After balancing using SMOTE, the target classes transformed into equal instances of 3596 for each class. Also, another recently developed in the year 2003 most efficient oversampling technique combined with SMOTE and Tomek's Link into a SMOTETomek is applied to balance the target class. This technique can trade off the minority and majority class imbalance by removing the duplicate data points. It can also help balance the dataset without hampering the integrity of the original data and control overfitting or underfitting issues of the neural network model. The capacity to create synthetic data for the minority class using SMOTE and the ability to delete data from the majority class that is recognized as a Tomek link, that is, samples of data from the majority class that are most similar to the data from the minority class. Due to these two strengths of this technique, we applied the SMOTETomek oversampling method. After applying SMOTETomek, the target class instances reached 3568 in each class. The syntactic data sample is created using the following mathematical calculation. Let x_i be a minority sample and the nearest chosen neighbor is x_{ik}. If $x_{ik1}, x_{ik2}, \ldots, x_{ikn}$ are the neighbors of *m*, the syntactic values x_{smote} will be generated by the formula in equation. 19.

$$x_{\text{smote}} = x_i + \delta \cdot (x_{ik} - x_i) \tag{19}$$

where, δ is a random number within the range of [0, 1], determining the position of the synthetic sample along the line segment connecting x_i and x_{ik}. The process will be repeated until the desired class balance is obtained.

$$d(x_i, x_j) < d(x_i, x_k) \tag{20}$$

Also, by using the Tomek link between the two pairs of instances (x_i, x_j), where x_i and x_j belong to different classes and are nearest to each other, the overlapping instances can be cleaned up after SMOTE. The Tomek link of pair (x_i, x_j) can be formed using the Eq. (20) for all other x_k, where the $d(x_i, x_j)$ is the distance between the two points and the x_k is the other instance.

Moreover, the most recently developed in the year 2004 another variation of the sampling technique SMOTE, SMOTEENN combined with SMOTE and Edited Nearest Neighbor (ENN) which works on the principal concept of KNN is implemented to manage the target class data imbalance. The SMOTE generally produces synthetic samples that are noisy and irrelevant in feature space regions where the minority class is highly concentrated. This may lead to overfitting and a decline in the model's capacity for generalization. The ENN approach, which

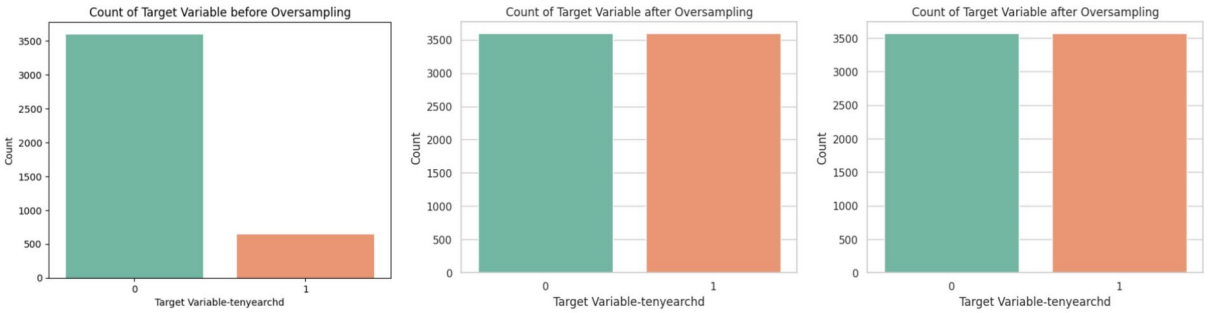


Fig. 5. The target class data distribution.

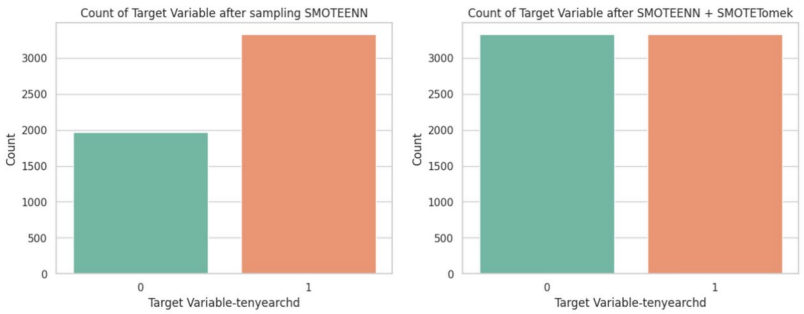


Fig. 6. The target class data distribution.

Target class level tenyearchd (yes/no)	# Before sampling	# After sampling			
		SMOTE	SMOTETomek	SMOTEENN	SMOTEENN+SMOTETomek
0	3596	3596	3568	1971	3326
1	644	3596	3568	3326	3326
Total #	4240	7192	7136	5297	6652

Table 2. The effect of sampling techniques on the target class data instances.

has data filtering capabilities that eliminate noisy and irrelevant samples from the dataset, can be combined with it to remedy this shortcoming. The ability of SMOTE can produce artificial examples for the minority class, while ENN can remove certain observations from both classes if it can determine that the observation's class differs from its K-nearest neighbor majority class. Therefore, by fusing SMOTE with ENN, SMOTENN can produce artificial samples that are less noisy and more representative of the minority class. This might enhance the model's ability to generalize.

So, finally after applying the SMOTEENN, the minority class is oversampled with the help of the SMOTETomek technique. So, the combined effect of SMOTEENN and SMOTETomek creates high-quality data by removing noise through ENN and refining the class boundary through the Tomek link. The effect of the two-stage oversampling and undersampling technique combined with SMOTEENN followed by SMOTETomek will lead to a highly cleaned and well-separated dataset.

Before and after the application of the sampling techniques SMOTE, SMOTETomek, SMOTEENN, and SMOTEENN + SMOTETomek, the data distribution of the target class is depicted in Figs. 5 and 6. In addition, the target class instances in class 1 and class 0 after applying the sampling techniques are represented for better understanding in Table 2.

After applying the required data cleaning process to the collected raw dataset, the dataset is now ready for the next model deployment. To train and test the neural network model using different optimizers, it is necessary to split the entire dataset into training and testing data. In each case, before and after applying the SMOTE, SMOTETomek, and SMOTEENN, the entire dataset is split into a ratio of 0.90:0.10 to train and test the data.

Proposed ANN-based prediction model

This is one of the principal stages in the whole working process and holds the highest importance in this research study of predicting the ten-year risk of CHD in a person. The ANN-based prediction model is being deployed in multiple stages of Dense layers and these stages parameters are been given below in detail. The details of the customized deployed model framework after applying preprocessing of the collected dataset feed into the neural network model are depicted in Fig. 7.

Model description

As already stated, it is the foremost approach used to train the model. It is not only a complex technique but also requires many packages to be imported before being employed on the dataset. The chief package involved in ANN's set-up is the "Keras" and "TensorFlow" library as it contains all the necessary components involved for neural network model deployment. Next, the "Sequential()" API is used to initialize the building of the neural network model and create the stacking layers. Then it is necessary to form the multiple nodes of the network within the dense layers while each layer has an activation function. The data are passed through the Input dense layers and, using the multiple hidden dense layers, the neurons receive the input from the neurons of the previous layer. The dense layer is a fully connected network and it ends with an output layer. Also, there should be dropout layers with a dropout value which acts as a filter for having a value well qualified for better training a node. In each step during training, the frequency of the dropout rate should be scaled up by 1 to control the overfitting of the model.

In the proposed architecture, the dense layers establish complete connections between all neurons in consecutive layers, facilitating the network's ability to understand intricate, hierarchical patterns within the data for deep feature extraction. The utilization of four dense layers enables the neural network to discern progressively intricate patterns and hierarchical associations within the input. This depth achieves a balance between model complexity and computing economy, allowing the network to learn intricate features without significant overfitting, hence enhancing generalization while minimizing unnecessary computational expense.

Also, dropout layers randomly deactivate neurons during training, diminishing dependence on particular neurons and mitigating overfitting. An elevated dropout rate starting with 0.5 in initial layers eliminates a greater number of connections, promoting diversified feature acquisition. Reduced dropout rates with 0.2 in deeper layers maintain acquired patterns while simultaneously regularizing the model. Utilizing diverse dropout rates to optimize regularization across layers, managing overfitting while preserving sufficient feature retention in deeper layers.

The choice of activation functions like ReLU effectively addresses non-linear interactions by activating solely positive inputs and inhibiting negative ones, thus avoiding vanishing gradients and facilitating expedited training. Whereas, the LeakyReLU mitigates the "dying neuron" problem associated with ReLU by permitting a minimal gradient for negative inputs, hence enhancing learning in contexts where features may exhibit negative correlations. So, combining ReLU and LeakyReLU optimizes performance and resilience, harnessing ReLU's efficiency while addressing its shortcomings. This design architecture promotes strong feature learning, rapid regularization, and adept management of non-linear data, hence ensuring the model's generalizability and precision.

In each dense layer including the input and hidden layers, the two kinds of activation functions ReLU and LeakyReLU were considered to get the output of each layer. The target class is in binary and due to the binary classification task, one neuron was used in the output layer with the "Sigmoid" activation function instead of "Softmax" to fetch the performance metrics as binary cross-entropy as a loss function.

Next, after deciding the structure of the model, the "compile()" function is used to compile the model. During the compilation, an optimizer is selected to update the weights based on the loss function and gradient. The binary cross-entropy is also used as a loss function which will measure the difference between the actual and the predicted outcomes with the metric being accuracy. This process is repeated multiple times each with a variation

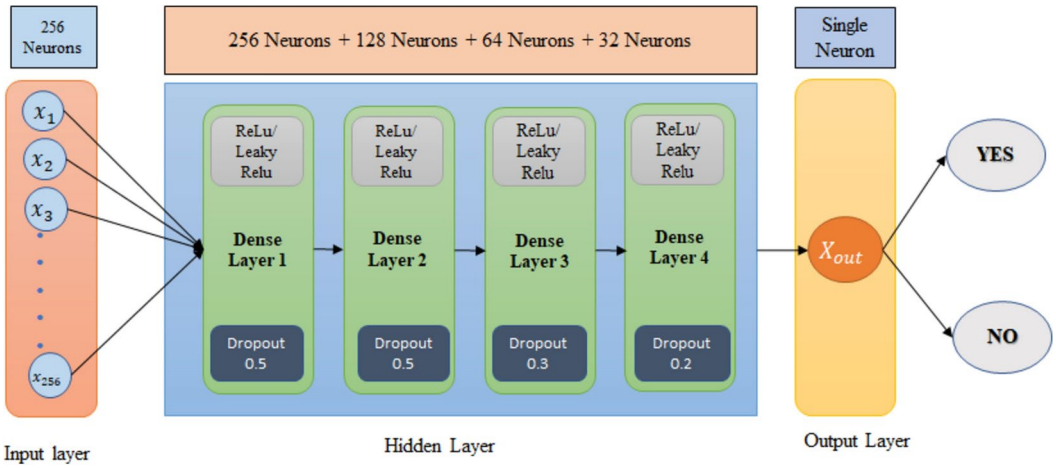


Fig. 7. The deployed artificial neural network model of the proposed methodology.

either in activation function or optimization technique. The ANN model employs activation functions such as ReLU and LeakyReLU to achieve better accuracy. The optimizer functions used for the improvement in accuracy are Adam, RMSProp, SGD, and AdaDelta to minimize the loss function during training. Also, the different learning rates are used to update the weights after calculating the gradient during the backpropagation process. The learning rate for each optimizer varies between the range of 0.01 to 0.0001 for getting the optimized value.

Finally, the model is trained using the "fit()" function on the supplied dataset. It carries out the actual learning process, in which the model uses the optimizer over several epochs to change its weights based on the data and the loss function. For the model, the sample size for training and prediction of the epoch is put to work with values 35, 50, and 100. The epoch values help in making the model work with a small and large sample size to improve the variations which improve the accuracy immensely. The number of training samples to process before updating the model's weights, 32, determines the batch size. Before going on to the next batch, the model processes 32 samples, calculates the loss, and modifies the weights depending on these samples. The smaller batch size is considered for frequent weight updates and a steady learning process. The entire flow of the model is represented through the algorithms in 1.

Step 1: Define the model architecture
Initialize Sequential model
Add Dense layer with 256 neurons, ReLU / LeakyReLU activation, and input shape $(input_dim)$
Add Dense layer with 256 neurons and ReLU / LeakyReLU activation
Add Dropout layer with rate 0.5
Add Dense layer with 128 neurons and ReLU / LeakyReLU activation
Add Dropout layer with rate 0.5
Add Dense layer with 64 neurons and ReLU / LeakyReLU activation
Add Dropout layer with rate 0.3
Add Dense layer with 32 neurons and ReLU / LeakyReLU activation
Add Dropout layer with rate 0.2
Add Dense layer with 1 neuron and "Sigmoid" activation **# for binary classification**
Step 2: Compile the model: "compile ()"
Compile model with:
    Optimizer = Adam(learning_rate=0.001) **or** SGD **or** RMSProp **or** AdaDelta
    Loss = Binary Cross-entropy **# For binary classification**
    Metrics = Accuracy
Display model summary: "sequential": model. summary ()
Step 3: Train the model: "fit ()"
Fit model on training data (X_{train}, y_{train}) **# With and without oversampling (SMOTE/SMOTETomek)**
Parameters:
    Epochs = 100
    Batch size = 32
    Validation data = (X_{test}, y_{test})
Step 4: Evaluate the model: "evaluate ()"
Evaluate model on training data (X_{train}, y_{train})
Evaluate model on test data (X_{test}, y_{test})
Step 5: Make predictions
Predict using the trained model on new input data (new_data)

Algorithm 1. Define, Compile, Train, Evaluate, and Predict with Artificial Neural Network Model.

The model is trained with a batch of several epochs to find a training score and accuracy. The training of the model with multiple parts of the data helps in improving the accuracy of the data by providing different data each time. The split of the training and testing is useful here as it allows the model to train and predict from different sections of the same dataset.

Performance metrics

To evaluate the model performances during training and evaluation, the binary cross-entropy metrics are employed to track the model's performance. Accuracy, which expresses the percentage of accurate predictions, is the most widely used indicator for classification issues. Nevertheless, other metrics can be added based on the assignment. The binary cross-entropy is calculated in the form of training accuracy and loss, and also validation accuracy and loss function. For better evaluation of the model, not only accuracy score is obligatory but also the multiple scores are needed to build an effectively precise model. The model deployed needs to be evaluated with the help of certain parameters which are required to be understood if it is not overfit or underfit with the given data. Binary Cross-entropy is the method employed for refined processing of the data with improved accuracy. The application of categorical cross entropy is avoided as it is simply observable that the data only contains two labels. Hence it is necessary to employ the binary cross-entropy with multi-label classification.

Such parameterized can not only improve the assumptions employed by the model for prediction but also provide standard metrics. The values achieved from accuracy present the correctness of the model but the precision values obtained show the model's ability to predict values nearest to the accurate value. By using the confusion matrix of each deployed model, Precision, Recall, and F1-Score were evaluated with the test dataset.

	Predicted Negative CHD = 0	Predicted Positive CHD = 1
Actual Negative CHD = 0	True Negatives (TN)	False Positives (FP) Type-I error
Actual Positive CHD = 1	False Negatives (FN) Type-II error	True Positives (TP)

Table 3. The confusion matrix concerning the CHD prediction model.

AF	E	TA (%)	TL	VA (%)	VL
ReLU	35	85.22	0.3692	85.38	0.3864
	60	86.27	0.3491	85.85	0.3953
	100	86.84	0.3211	85.61	0.4069
Leaky-ReLU	35	85.32	0.3692	86.08	0.3797
	60	85.88	0.3595	86.08	0.3857
	100	87.00	0.3287	86.79	0.3875

Table 4. The experimental outcome before oversampling using Adam optimizer (lr = 0.01).

In this case, the ratio of accurately predicted positive observations to the total number of predicted positives is represented as precision. High precision indicates a high likelihood of accuracy with few false positives during the positive prediction. Also, similarly, the ratio of accurately predicted positive observations to all actual positive observations is calculated by recall. In this regard, the high recall accurately detects the majority of positive cases with few false negatives. The recall plays a significant role in disease detection because missing a positive case is more important. The harmonic mean of recall and precision is represented by the F1 score. This is considered to identify a trade-off between precision and recall and having a highly unbalanced dataset.

During model testing, we derive the confusion matrix on the test dataset to evaluate the model's performance. In classification tasks, a confusion matrix serves as a performance evaluation tool, assessing the accuracy of a model's predictions against the actual results. It compares the predicted labels against the actual labels, offering insights into the nature of the errors committed by the model. It offers a comprehensive analysis of the model's accurate and erroneous predictions across various categories. The component of the confusion matrix is depicted in Table 3 to analyze the CHD model performance in this regard. The confusion matrix is essential for comprehending the nature of the model's errors, facilitating enhancements in performance, especially in imbalanced datasets when distinguishing between false positives and false negatives is critical. In the matrix, True Positives (TP): Accurately identified positive cases. True Negatives (TN): Accurately identified negative cases. False Positives (FP): Incorrectly classified as positive when they are actually negative (Type I mistake). False Negatives (FN): Incorrectly classified as negative when they are actually positive (Type II mistake).

Experimental setup

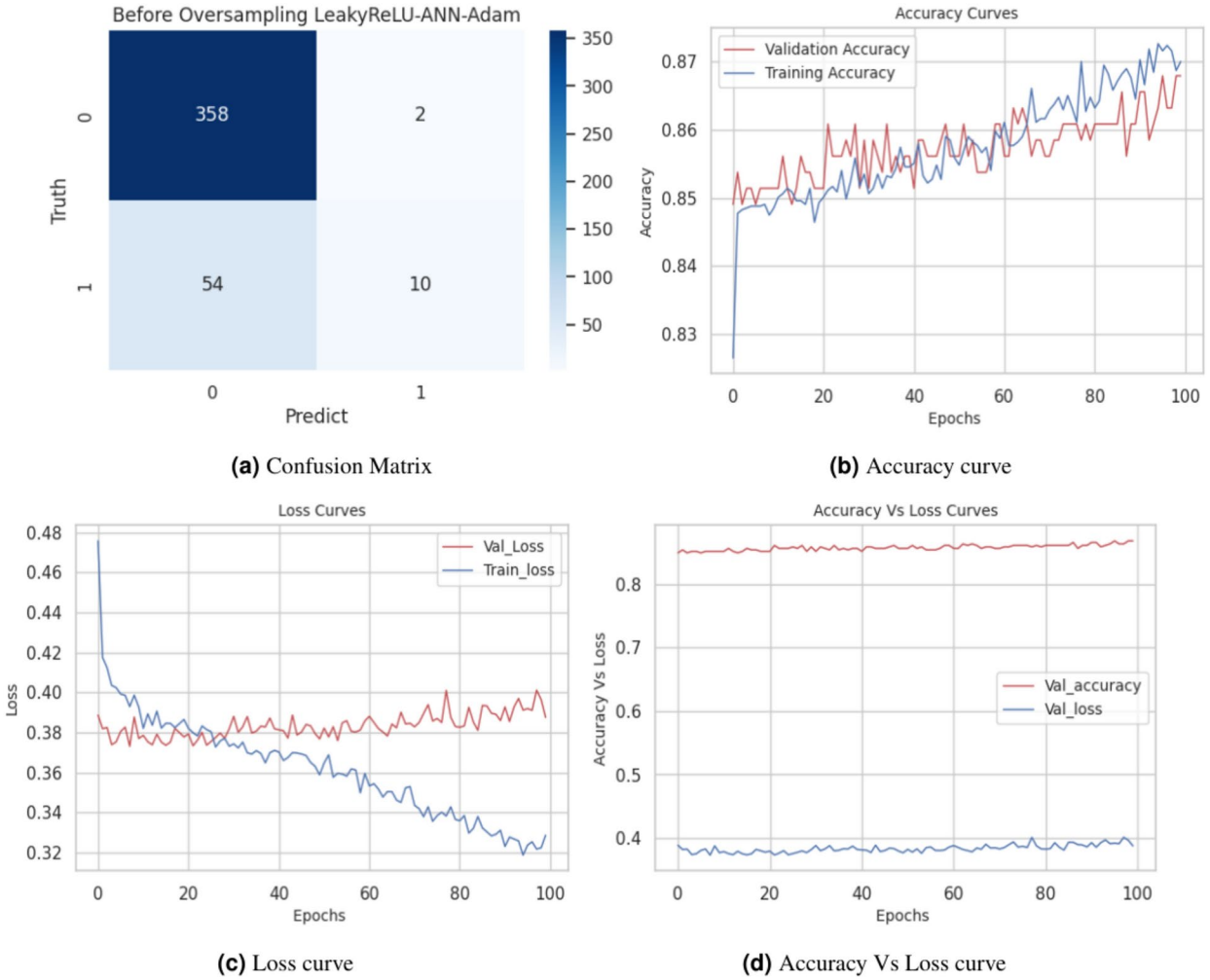
The experiment was conducted using the Anaconda environment setup to conduct the tests using the Python language. High performance is achieved by implementing the suggested model with an i5 processor, NVidia GeForce GTX GPU, and Windows 10 operating system. Creating a model that offers the NumPy library, which is needed to compute picture arrays, as well as the Anaconda Navigator and Shell for setting up the backend environment and installing several crucial supporting libraries. The transfer function and activation function are used to generate hidden layers in the ANN model, which is designed using Keras, an open-source neural network library that operates on top of TensorFlow. To make Keras more user-friendly, TensorFlow offers both high-level and low-level API.

Experimental result

This section demonstrates the outcome of the deployed artificial neural network after applying the data preprocessing techniques. To control the overfitting and underfitting of the model, the experiment was conducted with multiple parameters like batch size equal to 32, epochs run up to 100 and the dropout controlled with different rates in each layer ranging from 0.2 to 0.5.

The initial prediction model was deployed before the target class balancing after appropriate data cleaning methods. The Adam optimizer with a learning rate (lr) equal to 0.01 considering the activation function (AF) of ReLU and LeakyReLU used to achieve the output depicted in Table 4. The deployed ANN model considering the Adam optimizer with LeakyReLU performs better with a validation accuracy (VA) of 86.79%. The corresponding confusion matrix (CM), the training Accuracy (TA) versus validation accuracy, the training loss (TL) versus validation loss (VL) corresponding to epochs (E), and the Validation accuracy versus loss are represented in Fig. 8. The FN value is observed to be 54, which is 12.74% of the total text data in the CM.

In the second phase of model deployment on the preprocessed dataset after applying the initial oversampling technique using SMOTE, the target class was in equal instances of each class. A similar experimental setup is used to deploy the ANN model using different optimization techniques using Adam, SDG, RMSProp, and AdaDelta to achieve better accurate results. The optimization techniques are used to fine-tune the neural network parameters and architecture for better efficiency of the outcomes. The first Adam optimization approach outcome on the proposed ANN model for CHD prediction is depicted in Table 5. The sub-sequence optimization results



(a) Confusion Matrix **(b)** Accuracy curve

(c) Loss curve **(d)** Accuracy Vs Loss curve

Fig. 8. The performance curves of the ANN model before oversampling using LeakyReLU-Adam optimizer.

AF	E	TA (%)	TL	VA(%)	VL	Pr	Re	Fs
ReLU	35	98.28	0.0495	89.86	0.3578	0.88	0.96	0.93
	60	99.40	0.0190	91.25	0.3966	0.89	0.96	0.93
	100	99.98	0.0177	**93.75**	0.4905	0.91	0.97	0.94
LeakyReLU	35	98.01	0.0583	88.75	0.3177	0.86	0.96	0.91
	60	99.69	0.0148	90.97	0.3867	0.87	0.97	0.91
	100	100	0.0002	**92.36**	0.4165	0.92	0.96	0.92

Table 5. The Adam optimizer results after oversampling (SMOTE), (lr = 0.001). Significant values are in bold.

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	96.40	0.1001	88.75	0.3519	0.87	0.92	0.91
	60	100	0.0236	92.50	0.4319	0.88	0.96	0.92
	100	100	0.0132	92.50	0.5211	0.89	0.97	0.93
LeakyReLU	35	98.11	0.0538	86.94	0.3524	0.88	0.92	0.91
	60	99.51	0.0143	91.67	0.3682	0.88	0.95	0.91
	100	100	0.0003	92.50	0.4500	0.89	0.97	0.93

Table 6. The SDG optimizer results after oversampling (SMOTE), (lr = 0.001, momentum = 0.9).

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	98.39	0.0509	89.86	0.4025	0.86	0.91	0.88
	60	99.18	0.0274	90.83	0.5347	0.87	0.93	0.88
	100	99.64	0.0142	91.53	0.6969	0.88	0.96	0.91
LeakyReLU	35	97.87	0.0691	88.19	0.4754	0.85	0.90	0.89
	60	99.37	0.0284	90.97	0.5609	0.88	0.91	0.91
	100	99.57	0.0218	92.08	0.6241	0.90	0.94	0.92

Table 7. The RMSProp optimizer outcomes after oversampling (SMOTE), (lr = 0.01, rho = 0.9).

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	65.54	0.6560	66.11	0.6528	0.67	0.69	0.68
	60	66.30	0.6262	66.25	0.6235	0.67	0.69	0.68
	100	67.86	0.6032	66.81	0.6047	0.67	0.69	0.68
LeakyReLU	35	66.67	0.6462	64.31	0.6521	0.64	0.66	0.66
	60	67.26	0.6297	65.28	0.6382	0.64	0.67	0.66
	100	68.68	0.6013	66.25	0.6182	0.66	0.68	0.67

Table 8. The AdaDelta optimizer results after oversampling (SMOTE), (lr = 0.001, rho = 0.95).

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	98.65	0.0395	91.74	0.3914	0.91	0.97	0.93
	50	99.49	0.0181	92.86	0.3569	0.93	0.97	0.94
	100	100	0.0000	**93.98**	0.5001	0.94	0.97	0.94
LeakyReLU	35	98.56	0.0426	91.46	0.3245	0.91	0.96	0.94
	50	98.58	0.0366	92.02	0.3932	0.92	0.96	0.94
	100	100	0.0000	93.70	0.5084	0.93	0.97	0.94

Table 9. The Adam optimizer results after sampling (SMOTETomek), (lr = 0.001). Significant values are in bold.

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	99.92	0.0075	93.96	0.3197	0.90	0.98	0.95
	50	100	0.0000	94.34	0.5481	0.91	0.98	0.96
	100	100	0.0000	**94.53**	0.7601	0.92	0.99	0.96
LeakyReLU	35	98.76	0.0484	92.82	0.3335	0.90	0.97	0.93
	50	99.63	0.0108	93.77	0.3176	0.92	0.99	0.95
	100	100	0.0000	94.34	0.7002	0.92	0.99	0.95

Table 10. The Adam optimizer results after sampling (SMOTEENN), (lr = 0.001). Significant values are in bold.

concerning SDG, RMSProp, and AdaDelta with the same experimental environments are represented in Tables 6, 7, and 8, respectively.

Also, after observing the experimental outcomes in the second phase of model deployment using the oversampling technique SMOTE, the Adam optimizer performed better than other optimizers. So, we decided to apply the most recently developed oversampling technique SMOTETomek on the highly unbalanced dataset to remove the redundant instances from the majority class samples and to trade off between the minority and majority classes. This technique is used to verify the significance of using it to tackle the highly unbalanced target class. Also, lastly developed sampling technique SMOTEENN, a variation of SMOTE with Edited Nearest neighbor is chosen to remove the noise from the majority class and handle the minority class properly. These techniques are used to verify the significance of usage to handle the highly unbalanced target class data instances and also compare the superiority of those techniques. Similar experimental environments were considered to deploy the ANN model using the best-performing Adam optimizer with the same parameters as before used in the case of SMOTE along with Adam. The experimental results were observed slightly high accuracy of 93.98% and 94.53% corresponding to ReLU along with Adam after the sampling effect of SMOTETomek and SMOTEENN.

The corresponding results are represented in Tables 9 and 10 respectively for a better understanding of their insight overview.

Finally, after careful observation of the deployed model performance with the effect of SMOTEENN and SMOTETomek, we decided to apply the two-stage sampling approach to combine the sequential effect of SMOTEENN and SMOTETomek to combat the minority class issue and also to separate the two classes with the help of a prominent decision boundary. The combined effect demonstrated the highest accuracy of 96.25% coupled with the Adam-ReLU combination. The observable outcomes are represented in Table 11.

Analysis and discussion

The best-performing optimization technique using the Adam optimizer after the target class oversampling effect of using SMOTE with the ReLU activation function achieved an accuracy of 93.75%. The validation loss was moderate at around 0.4905, and the other metrics, like recall, were also high, indicating the prediction stability with 0.97. The predicted accuracy vs. loss during training and validation concerning epochs and the confusion matrix (CM) for binary classification is depicted in subsequent figures. The corresponding CM, the training versus validation accuracy, the training versus validation loss corresponding to epochs, and the validation accuracy versus loss are represented in Fig. 9. The corresponding FN value was significantly decreased in this case, as observed with 10, which is 1.39% of the test data.

Also, the effect of the other applied sampling techniques like SMOTETomek and SMOTEENN performed well with the Adam optimizer considering the ReLU activation function. The observable validation accuracy was 93.98% and 94.53% with the recall values of 0.97 and 0.99, respectively. The corresponding performance curves, like those of other models of SMOTETomek and SMOTEENN, are represented in Figs. 10 and 11, respectively. The FN rates were reasonably accounted for with respect to SMOTETomek and SMOTEENN at 1.26% and 0.56%, respectively.

Moreover, after the sequential effect of SMOTEENN and SMOTETomek using the ReLU activation function with Adam optimizer performed better than the previous all deployed models compared with the after effect of SMOTETomek and SMOTEENN alone. The combined model performance was achieved with an accuracy of 96.25% with a loss function of 0.5364. Moreover, the LeakyReLU reached slightly less accuracy compared to ReLU but both of them are highly efficient predictive models with recall values of 0.98 each. This indicates the stability and effectiveness of the proposed methods for CHD disease prediction on highly unbalanced medical datasets. Due to the high recall values, the effective false negative (FN) rate was 0.75%, less than 1% indicates 99.25% correctly predicted positive observations as positive. Similar to SMOTE, the accuracy vs loss curve during the training and validation considering 100 epochs, CM, and validation accuracy vs loss curves is depicted in Fig. 12 with the effect of SMOTEENN+SMOTETomek.

The utilization of those optimizers displays that the model is based on recently developed algorithms. Based on multiple gradient descent techniques, it manifests an astonishing accuracy when deployed with LeakyReLu. This result shows that the new optimizer libraries are very helpful in developing the CHD prediction model. Adam Optimizer improves the accuracy of the ANN model and the model's fitting, which helps provide more stable results. The outcomes achieved from the optimizer SGD and RMSProp are only the nearest to the final proposed results as they work with a similar mechanism. In contrast, the AdaDelta differs with a much higher gap.

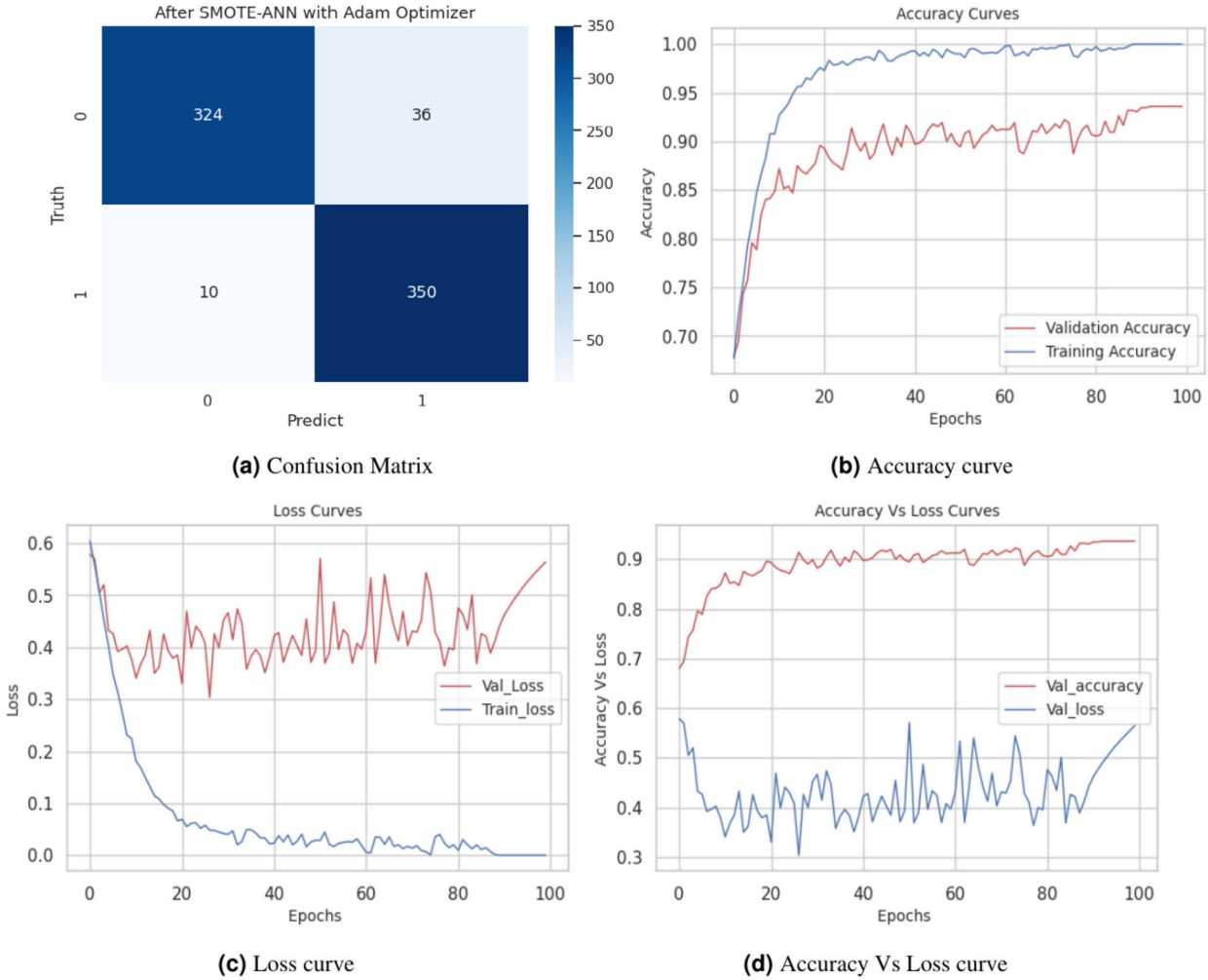
Also, the results achieved using two activation functions show the best result in the multiple tests performed with many different parametrics. Each parameter on tuning exhibits a change in the outcome but still, the function prevails with all the technicalities. LeakyReLU is the other activation function for the training and it is easily observable that it is close to the results procured by ReLU and in some cases even gets better than it. This is evidence of the fact that a similar working technique that uses a similar implementation can only make the accuracy better.

The comparison of the considered optimizers concerning activation functions' experimental performance in terms of validation accuracy and loss function is plotted in Fig. 13 after the impact of the SMOTE technique. The insight views of the optimizers are significant for the prediction of coronary heart disease in neural networks showing important factors.

Also, the influence of the sampling techniques SMOTE, SMOTETomek, SMOTEENN, and SMOTEENN+SMOTETomek is impacted heavily after applying the best performing Adam optimizer concerning ReLU and LeakyReLU activation functions. The performance comparison of sampling approaches is depicted in Fig. 13 for insight and understanding of the effect.

AF	E	TA (%)	TL	VA (%)	VL	Pr	Re	Fs
ReLU	35	99.32	0.0189	94.59	0.3017	0.92	0.97	0.95
	50	100	0.0000	96.10	0.4212	0.94	0.98	0.96
	100	100	0.0000	**96.25**	0.4364	0.94	0.98	0.96
LeakyReLU	35	99.06	0.0282	94.44	0.2751	0.91	0.97	0.93
	50	100	0.0000	95.20	0.3371	0.93	0.98	0.95
	100	100	0.0000	95.35	0.5005	0.93	0.98	0.95

Table 11. The Adam optimizer results after sampling (SMOTEENN+SMOTETomek), (lr = 0.001). Significant values are in bold.



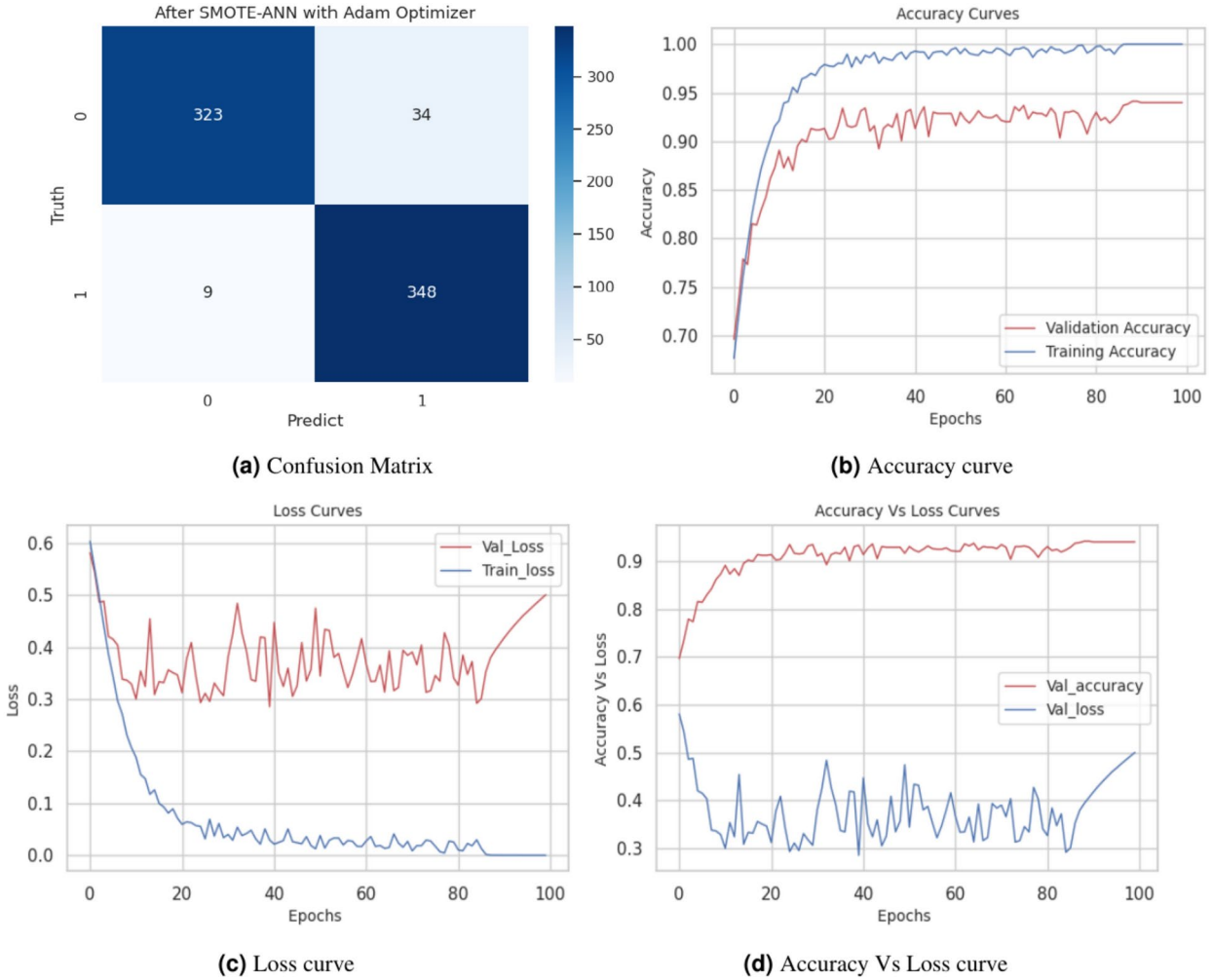
(a) Confusion Matrix **(b)** Accuracy curve

(c) Loss curve **(d)** Accuracy Vs Loss curve

Fig. 9. The performance curves of the proposed ANN model after oversampling (SMOTE) with Adam optimizer and ReLU activation function.

Comparison of experimental results with the SOTA learning techniques

The Framingham Heart Study (FHS) dataset is used in this article to compare the experimental results with the other proposal. We sum up the findings of the prior literature review by considering accuracy during validation. To improve accuracy before model deployment, a small number of articles employed feature selection and elimination techniques. This research does not use any feature elimination technique to reach higher efficiency but rather to control the overfitting risk by tuning the model's parameters. As shown in Table 12, the suggested work yields more accurate results when compared to the most recent studies utilizing the FHS Dataset. Few research articles in reputed journals concentrate their study on deploying the coronary heart disease prediction model using an artificial neural network. Rather uses the classical ML models and developed ensemble learning techniques for specific tasks. Few researchers use feature selection approaches with genetic algorithms but observable accuracy was not researched in the milestone. Also, the encoder-based approaches do not provide good results. This research approach highlights the effect of oversampling and optimization techniques in artificial neural networks for generalized disease prediction models. Due to the customized design of the neural network model, considering the four hidden dense layers with the optimal number of neurons in each layer helps to learn the intricate relationship among them. Also, the varying uses of dropout rates in each layer optimize the regularization across layers and manage overfitting while preserving sufficient feature retention in deeper layers. The combination of the appropriate use of the activation function in each layer and the selection to fit the best optimizer produces outperforming results. Those challenges were handled in a well-mannered way with the specific intention of optimizing the artificial neural network model for the prediction of CHD. Moreover, the well-managed data preprocessing approaches to handle the target class data imbalance by applying the advanced sampling techniques achieved the optimal outcomes. The proposed novel two-stage sampling approach, by considering the SMOTEENN and SMOTETomek and their sequential implication of combined effect, reached milestone experimental results with new heights. Given their ability to manage intricate non-linear relationships among input features and their automated feature extraction from raw datasets, ANN-based coronary heart disease prediction models will enhance efficacy in contemporary healthcare decision-making systems. Given



(a) Confusion Matrix **(b)** Accuracy curve

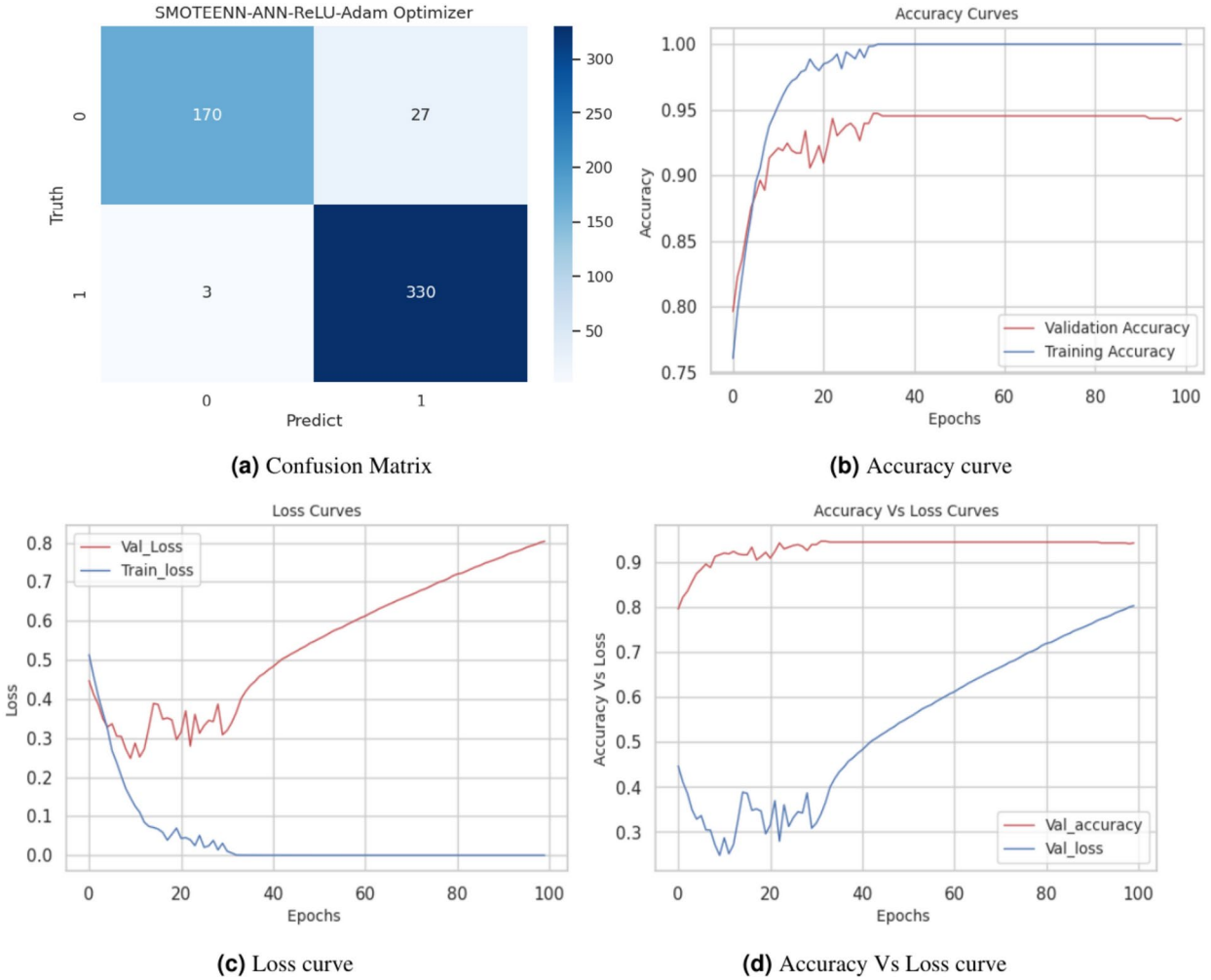
(c) Loss curve **(d)** Accuracy Vs Loss curve

Fig. 10. The performance curves of the proposed ANN model after oversampling (SMOTETomek) with Adam optimizer and ReLU activation function.

their ability to manage intricate non-linear relationships among input features and their automated feature extraction from raw datasets, ANN-based coronary heart disease prediction models will enhance efficacy in contemporary healthcare decision-making systems.

Conclusion

This study motivates us to design and develop a prediction system that is capable of detecting the risk of coronary heart disease in persons in the next ten years. The data mining technique based on a nature-inspired neural network is a better alternative for a quick detection support system compared to traditional machine learning classifiers. By using optimization techniques, the artificial neural network can optimize the parameters easily to achieve the highest optimal performance. This research considers the many neural network optimizers like Adam, SDG, RMSProp, and AdaDelta to verify the significance of their uses for network tuning to update weights, minimize loss functions, and also justify the outcomes. Also, the two types of activation functions ReLU and LeakyReLU are considered in dense layers to verify their effect in handling vanishing gradient and dead neuron problems. Moreover, the different parameters like learning rate, batch size, and epochs are optimized to control the underfitting and overfitting issues. The target class data imbalance is addressed by applying the oversampling techniques through SMOTE and other sampling methods like SMOTETomek, and SMOTEENN, and also with the proposed two-stage sampling approach combined with best performing SMOTEENN and SMOTETomek. In the end, the superior Adam optimizer with appropriate parameters, among others, coupled with the combining effect of SMOTEENN and SMOTETomek and ReLU reached the highest validation accuracy of 96.25% compared to the other approaches of outcomes. The validation loss of the proposed model was minimal, with high recall values of 0.98, which indicates an efficient coronary heart disease risk prediction model so far. This research study concludes with the new benchmark performance accuracy so far regardless of other proposed approaches for the prediction of coronary heart disease.



(a) Confusion Matrix **(b)** Accuracy curve

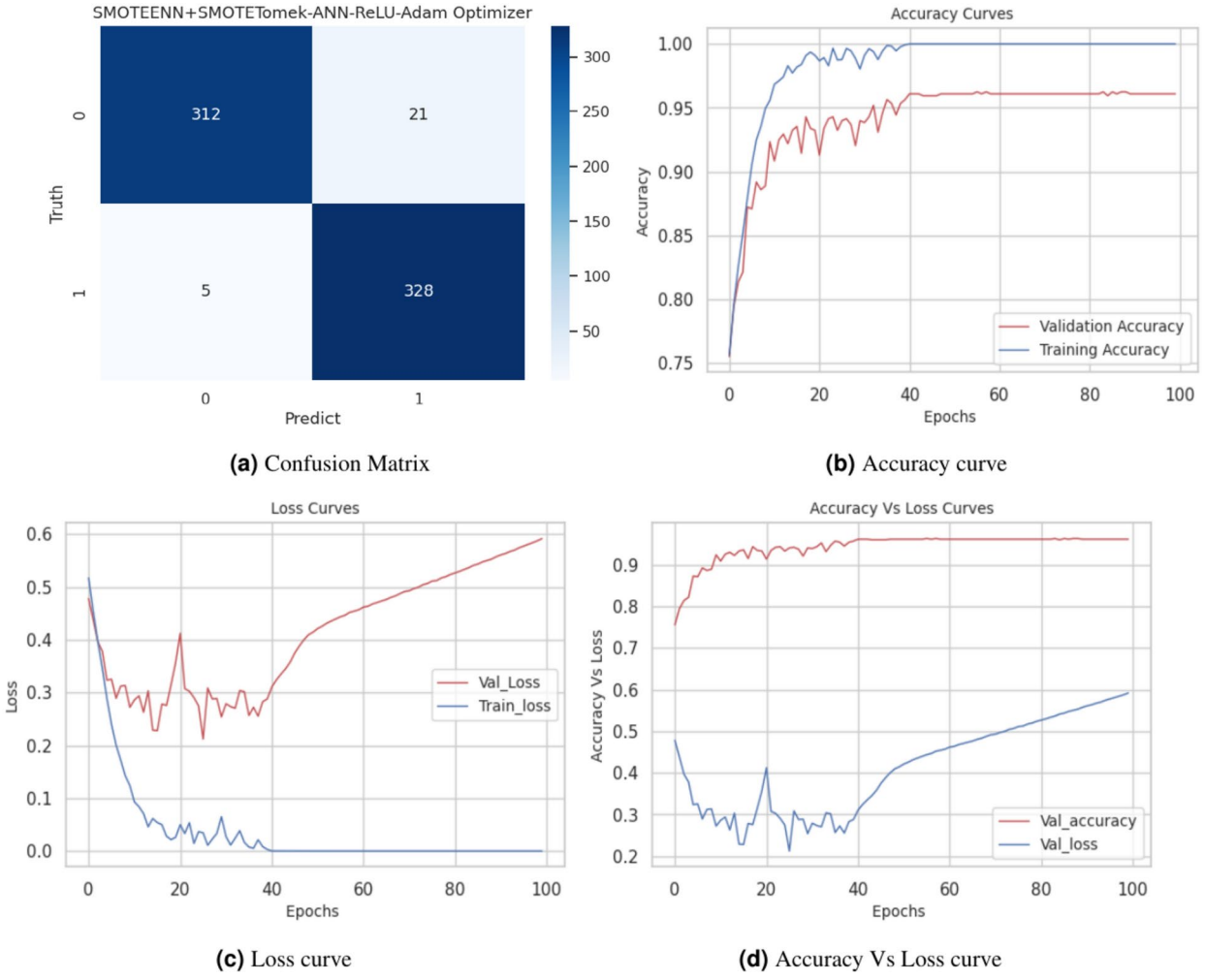
(c) Loss curve **(d)** Accuracy Vs Loss curve

Fig. 11. The performance curves of the proposed ANN model after oversampling (SMOTETENN) with Adam optimizer and ReLU activation function.

Limitations and future scope

This study's main challenge and limitation is collecting real-time modern healthcare patients' information and aggregating it into a healthcare application due to its unstructured nature and limited medical data resources. So, thorough data preprocessing is required before model development for disease prediction. Also, the single dataset with fewer instances was collected from an online data repository for research purposes. The team is working with the collection of real-time raw hospital datasets for further analysis and deployment of an application through a chatbot. Due to the sensitive medical data, it is likely prone to compromise the security and privacy of patients' data to unauthorized parties. These challenges can be addressed by adopting the federated learning approaches in a decentralized manner to secure data and achieve privacy easily.

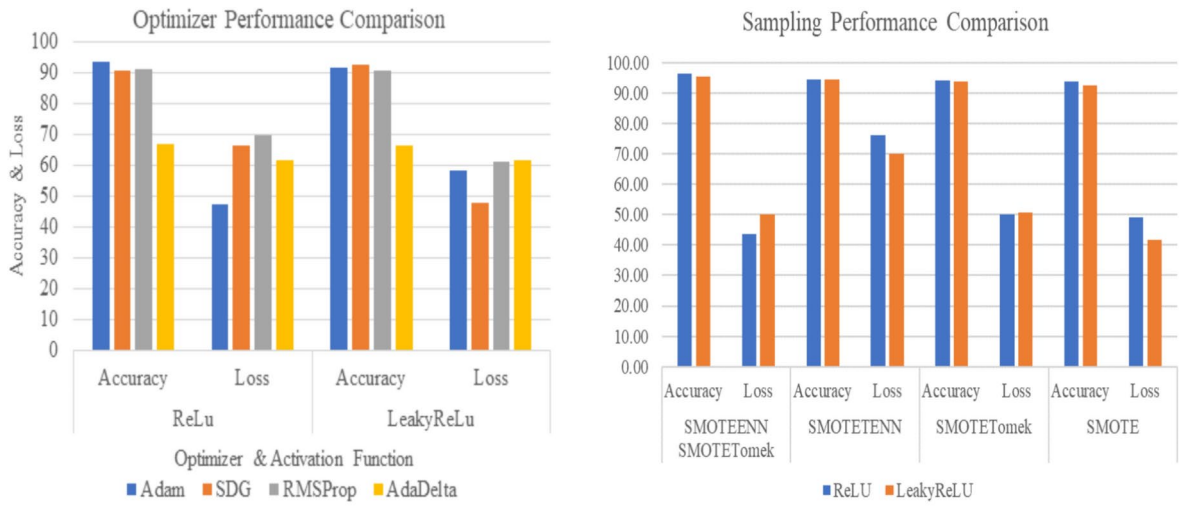
This study can be extended with diversified large datasets and nature-inspired evolutionary algorithms may eventually be combined with the emotional neural network for better outcomes. Several feature selection techniques can identify the key elements in the CHD diagnosis for more precise performances. Also, security and privacy can be achieved through cutting-edge federated learning techniques by implementing a distributed privacy-preserving manner.



(a) Confusion Matrix **(b)** Accuracy curve

(c) Loss curve **(d)** Accuracy Vs Loss curve

Fig. 12. The proposed ANN model's performance curves after sampling SMOTEENN+SMOTETomek with Adam optimizer and ReLU activation function.



(a) Optimizer Techniques Comparison **(b)** Sampling Techniques Comparison

Fig. 13. The effect of optimizers and sampling techniques performance comparison (Adam) concerning validation accuracy and loss function.

#Ref	Dataset name	Techniques used	Accuracy (%)
[33]	FHS	Boosted Decision Tree Classifier combined with R-Studio	85
[44]	FHS	RF without Feature selection	91.6
		RF with GA as a Feature selection	90.7
[43]	FSH	ANN based on a sparse autoencoder (SAE)	90
[26]	Hospital	RF Classifier (ML Model)	92.04
[29]	Hospital	KELM with salp swarm algorithm (SSA)	84.40
[42]	FSH	MLP based on a Neural Network model without a feature selection approach	92.87
[31]	FHS	RF with Recursive Feature Elimination (RFE) feature selection method	89
Proposed model		Customized ANN model with the combination of SMOTEENN+SMOTETomek-ReLU-Adam Optimizer	96.25

Table 12. Comparison of the proposed technique with the previous studies conducted on FHS.

Data availability

The dataset utilized in this research is publicly available in the open-source Kaggle repository (https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset). The entire deployment code is in our custody, it will be available whenever it is needed. The first author (Subhash Mondal, ph22cse1001@cit.ac.in) will provide deployment code, and data for further contact upon request.

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Author contributions
The author, Subhash Mondal conducted the Experiments, Conceptualization, Methodology, Writing the original draft, Writing the Review and Editing, and Formal analysis of the results; Also, the other authors, Ranjan Maity and Amitava Nag are responsible for the Conceptualization, Supervision, Investigation, Review and Editing, and Validation and analyzed the results. All authors read and reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Ethics approval
None of the authors of this article have conducted any research on humans or animals.

Additional information
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